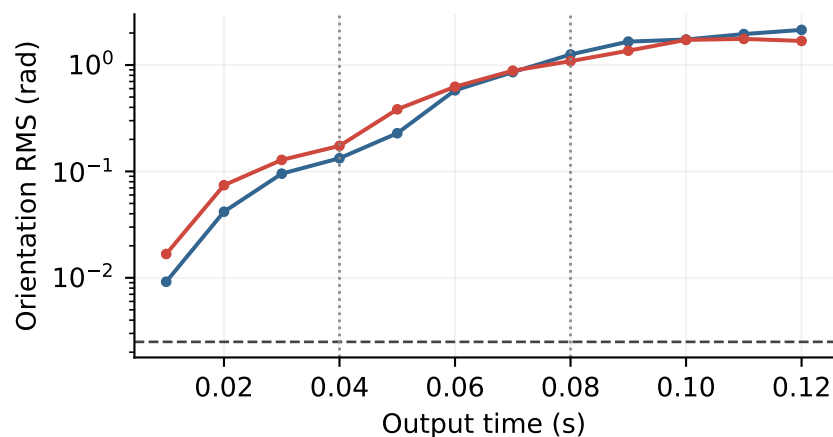
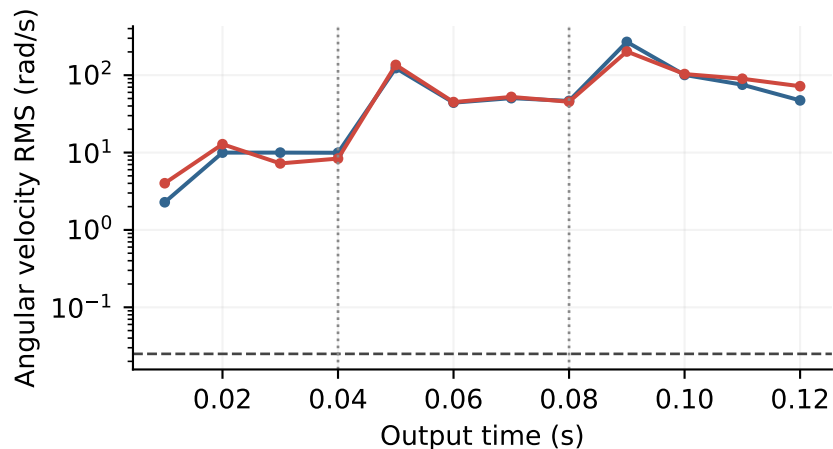
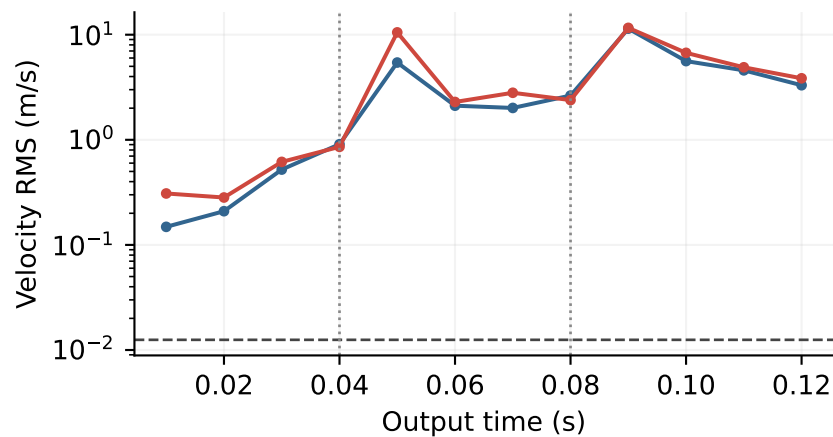
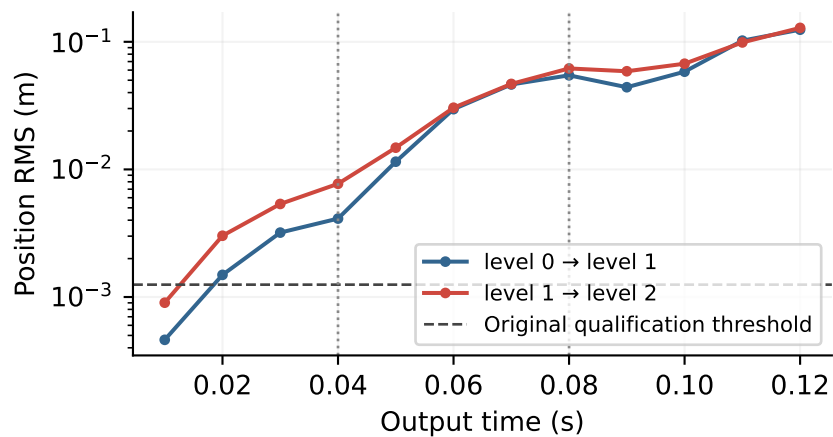


Random hulls: contact gates pass, trajectory refinement fails



Dotted lines: wall reversals. Divergence is already visible at 0.01 s, before either reversal.
Original source/scene/material preserved; no causal identification from output traces alone.